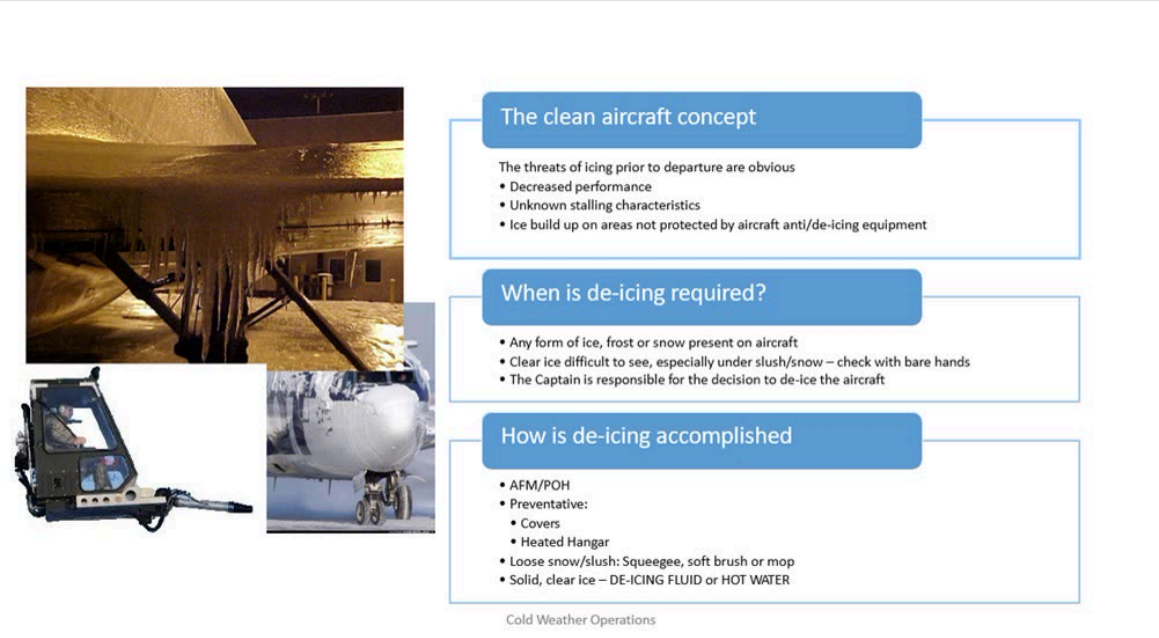
MODULE 3 · INSPECTIONS

# Lesson 13: The Pre-take-off Check and the Pre-take-off Contamination Check

Lesson 13 of 30 · about 8 min

**LEARNING OBJECTIVES**

- Distinguish the pre-take-off check from the pre-take-off contamination check (PTCC).
- State when each is required, who performs it and how.
- Recognise the signs of anti-icing fluid failure.




Before take-off, the question is always the same: are the critical surfaces clean?

## Two different checks

| | Pre-take-off check | Pre-take-off contamination check (PTCC) |
|---|---|---|
| When | Within the holdover time, as part of the before-take-off actions | When the holdover time has been **exceeded**, or when the crew is in doubt |
| Timing | Before take-off, as conditions require | Within **5 minutes** before beginning the take-off |
| Who / where | Flight crew, normally from inside the aircraft (wings or representative surfaces) | A qualified person, normally from **outside** the aircraft, or from inside only where an approved procedure allows it |
| What | The wings or representative surfaces that can be seen | The critical surfaces needed to confirm the aircraft is clean |
| If contamination is seen | Do not take off. Return for treatment. | |

A PTCC cannot be used to extend an **allowance time** (ice pellets, small hail) — see Lesson 8.

## Signs of fluid failure

- The surface loses its gloss and looks dull or matt;
- Snow builds up on the surface rather than being absorbed;
- Ice crystals or slush appear in the fluid;
- Frozen fluid or a "slushy" texture, often starting at the leading edge or wing tips;
- Accumulation on other surfaces, such as wiper blades or nose cone, while the HOT is still running (a useful early warning).

Fluid failure normally starts on the **leading edge and wing tips**, where the fluid layer is thinnest, and on surfaces exposed to wind.

## Practical points

- Use wing inspection lights at night. Check both sides.
- Cabin crew may be asked to look at the wings, using the company procedure. They must know what they are looking for and how to report it.
- If conditions get worse — heavier snow, a change to freezing rain — the HOT you calculated is no longer valid. Recalculate or return.

**DECISION POINT**

Agree before taxi: "If the HOT expires before take-off and a PTCC is not available, we return for treatment." A pre-briefed decision protects you from pressure at the holding point.

**KEY TAKEAWAYS**

- Pre-take-off check: within HOT, flight crew, usually from inside.
- PTCC: HOT exceeded, within 5 minutes of take-off, normally from outside the aircraft.
- Dull surface, snow building up or ice crystals in the fluid mean fluid failure — return for treatment.
- Pre-brief the decision to return before you taxi.

**KNOWLEDGE CHECK**

**The holdover time has expired while you queue for take-off. Snowfall continues. What is required to take off?**

A. Nothing, provided the wings look clean from the flight deck

B. A pre-take-off contamination check within 5 minutes before take-off, normally from outside — otherwise re-treat

C. Engine and wing anti-ice on

D. A new anti-icing code by radio

**Answer: B.** With HOT exceeded, take-off is only permitted after a PTCC, completed within 5 minutes before take-off, that confirms the critical surfaces are clean. Otherwise the aircraft must be re-treated.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.